



早稲田大学
WASEDA University

WASEDA University

Waseda Research Institute for Science and Engineering

Fundamentals of Programming

孫 鶴鳴

SUN Heming

Apr 14th



Self Introduction

◆ Education

❖ Bachelor: Shanghai JiaoTong University (2007-2011)

- Electronic Engineering



❖ Master: Waseda University (2010-2012)

- Integrated Systems



❖ Master: Shanghai JiaoTong University (2011-2014)

- Electronic Engineering



❖ Doctor: Waseda University (2014-2017)

- Integrated Systems



Self Introduction

◆ Career

❖ NEC central research lab (2017-2018)

- Researcher

❖ Waseda University (2018-2023)

- Junior Researcher (Assistant Professor)

❖ Yokohama National University (2023-2026)

- Associate Professor

❖ Institute of Science Tokyo (2026-now)

- Associate Professor

Orchestrating a brighter world

NEC



早稲田大学
WASEDA University

YNU YOKOHAMA
National University

 Institute of
SCIENCE TOKYO

Outline

- ◆ About Course
- ◆ What is Programming?
- ◆ Log-in Linux System
- ◆ Simple Commands of Linux System
- ◆ Exercise

About Course

◆ Concept

- ❖ Introduce the skills of C programming for beginners including the basic operations of variables, control flow, arrays, pointer, functions and so on.

◆ Target

- ❖ Help students to build confidence to write simple C programs.
- ❖ Help students to create simple applications by using C programs.

About Course

◆ Course time

❖ Spring quarter

- Tuesday: 10:40-12:20
- Friday: 8:50-10:30
- Face-to-face

◆ Grade Structure

- ❖ Exam: 60%
- ❖ Report and exercise: 20%
- ❖ Attendance: 20%

Waseda Moodle

Login

 WASEDA University

※ID必須

Login ID*

Password*

Login

Login IDにはWaseda IDを入力してください。
Waseda IDが不明な場合は、こちらから確認することができます。
(確認には、Waseda メールアドレスとパスワードの入力が必要です。)

> ログインできない場合
> ITサービスナビ

For "Login ID", please enter your Waseda ID.
If you forget your Waseda ID, you can confirm it at this webpage.
(Waseda mail address and password are required.)

> If you cannot log in.
> IT Service Navigation

授業支援 ▼

Waseda Moodle

成績入力

シラバス入力・閲覧

休講・補講

学生授業アンケート

非正規科目の新規設置申請 および科目への利用者追加・削除申請

Waseda Moodle に関するお問い合わせ

Teach Anywhere

ID : TAwaseda PW : Anywhere1021

Learn Anywhere

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Difference with Other Courses

Mathematics

$$(C)' = 0$$

$$(x^\mu)' = \mu x^{\mu-1}$$

$$(a^x)' = a^x \ln a$$

$$(e^x)' = e^x$$

$$(\log_a x)' = \frac{1}{x \ln a}$$

$$(\ln x)' = \frac{1}{x}$$

$$(\sin x)' = \cos x$$

$$(\cos x)' = -\sin x$$

$$(\tan x)' = \sec^2 x$$

$$(\cot x)' = -\csc^2 x$$

$$(\sec x)' = \sec x \tan x$$

$$(\csc x)' = -\csc x \cot x$$

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$$

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}}$$

$$(\arctan x)' = \frac{1}{1+x^2}$$

$$(\operatorname{arccot} x)' = -\frac{1}{1+x^2}$$

$$d(C) = 0$$

$$d(x^\mu) = \mu x^{\mu-1} dx$$

$$d(a^x) = a^x \ln a dx$$

$$d(e^x) = e^x dx$$

$$d(\log_a x) = \frac{1}{x \ln a} dx$$

$$d(\ln x) = \frac{1}{x} dx$$

$$d(\sin x) = \cos x dx$$

$$d(\cos x) = -\sin x dx$$

$$d(\tan x) = \sec^2 x dx$$

$$d(\cot x) = -\csc^2 x dx$$

$$d(\sec x) = \sec x \tan x dx$$

$$d(\csc x) = -\csc x \cot x dx$$

$$d(\arcsin x) = \frac{1}{\sqrt{1-x^2}} dx$$

$$d(\arccos x) = -\frac{1}{\sqrt{1-x^2}} dx$$

$$d(\arctan x) = \frac{1}{1+x^2} dx$$

$$d(\operatorname{arccot} x) = -\frac{1}{1+x^2} dx$$

$$\int k dx = kx + C$$

$$\int x^\mu dx = \frac{x^{\mu+1}}{\mu+1} + C$$

$$\int a^x dx = \frac{a^x}{\ln a} + C$$

$$\int e^x dx = e^x + C$$

$$\int \frac{dx}{x} = \ln |x| + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \frac{dx}{\cos^2 x} = \int \sec^2 x dx = \tan x + C$$

$$\int \frac{dx}{\sin^2 x} = \int \csc^2 x dx = -\cot x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \csc x \cot x dx = -\csc x + C$$

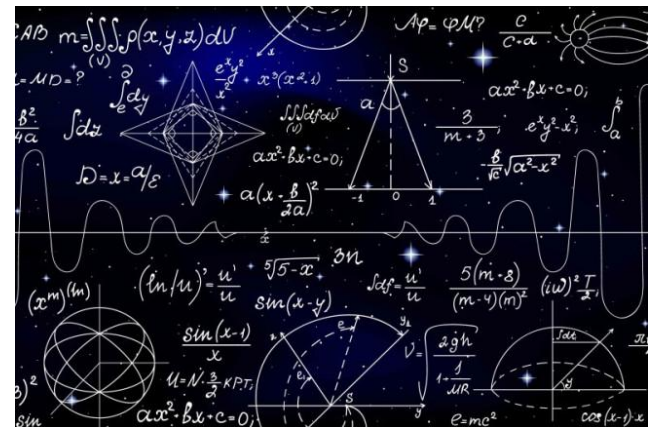
$$\int \frac{dx}{\sqrt{1-x^2}} = \arcsin x + C$$

$$\int \frac{dx}{1+x^2} = \arctan x + C$$

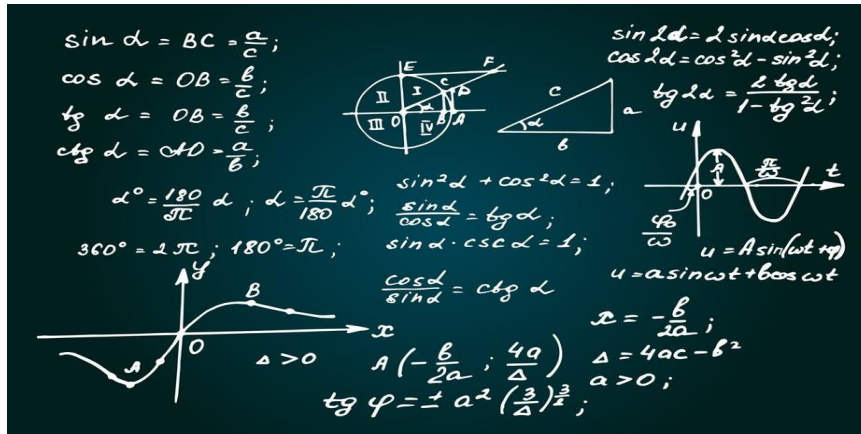
Chemical



Physics



Programming



Implement
everything on
the computer

Matrix Multiplication

The diagram illustrates the first step of matrix multiplication. It shows two matrices being multiplied. The first matrix is a 2x3 matrix with elements $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$. The second matrix is a 3x2 matrix with elements $\begin{bmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \end{bmatrix}$. A red curved arrow points from the first row of the first matrix (1, 2, 3) to the first column of the second matrix (10, 20, 30). The elements 1, 2, 3 and 10, 20, 30 are highlighted with a yellow oval, indicating the specific elements being multiplied together to calculate the first element of the resulting matrix.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \end{bmatrix}$$

Matrix Multiplication

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \end{bmatrix}$$

Diagram illustrating the dot product of the first row of the first matrix and the first column of the second matrix. Red arrows point from the elements 1, 2, and 3 of the first row to the elements 10, 20, and 30 of the first column.

$$= \begin{bmatrix} 1 \times 10 + 2 \times 20 + 3 \times 30 & 1 \times 11 + 2 \times 21 + 3 \times 31 \\ 4 \times 10 + 5 \times 20 + 6 \times 30 & 4 \times 11 + 5 \times 21 + 6 \times 31 \end{bmatrix}$$
$$= \begin{bmatrix} 10+40+90 & 11+42+93 \\ 40+100+180 & 44+105+186 \end{bmatrix} = \begin{bmatrix} 140 & 146 \\ 320 & 335 \end{bmatrix}$$

Some Basic Examples of C Programming

◆ Sorting algorithm

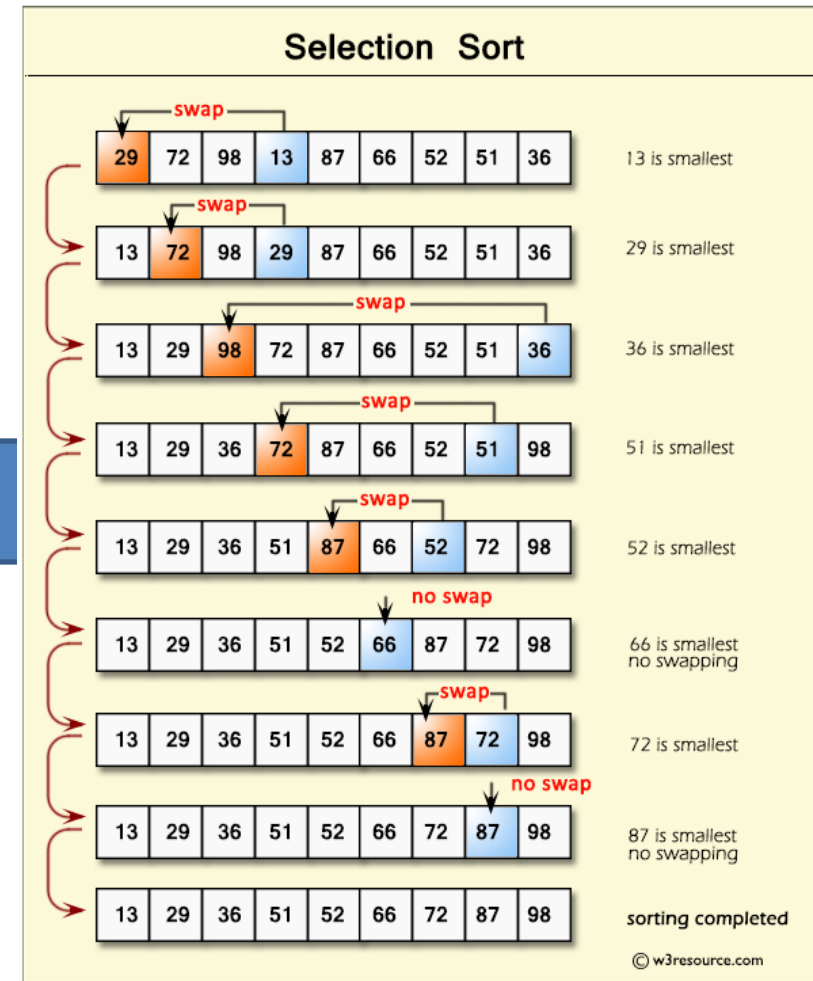
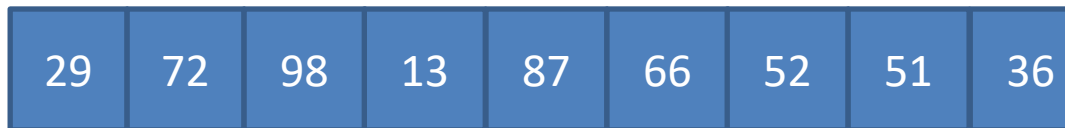
✦ Put elements of a list in an ascending order

29	72	98	13	87	66	52	51	36
----	----	----	----	----	----	----	----	----

Some Basic Examples of C Programming

◆ Sorting algorithm

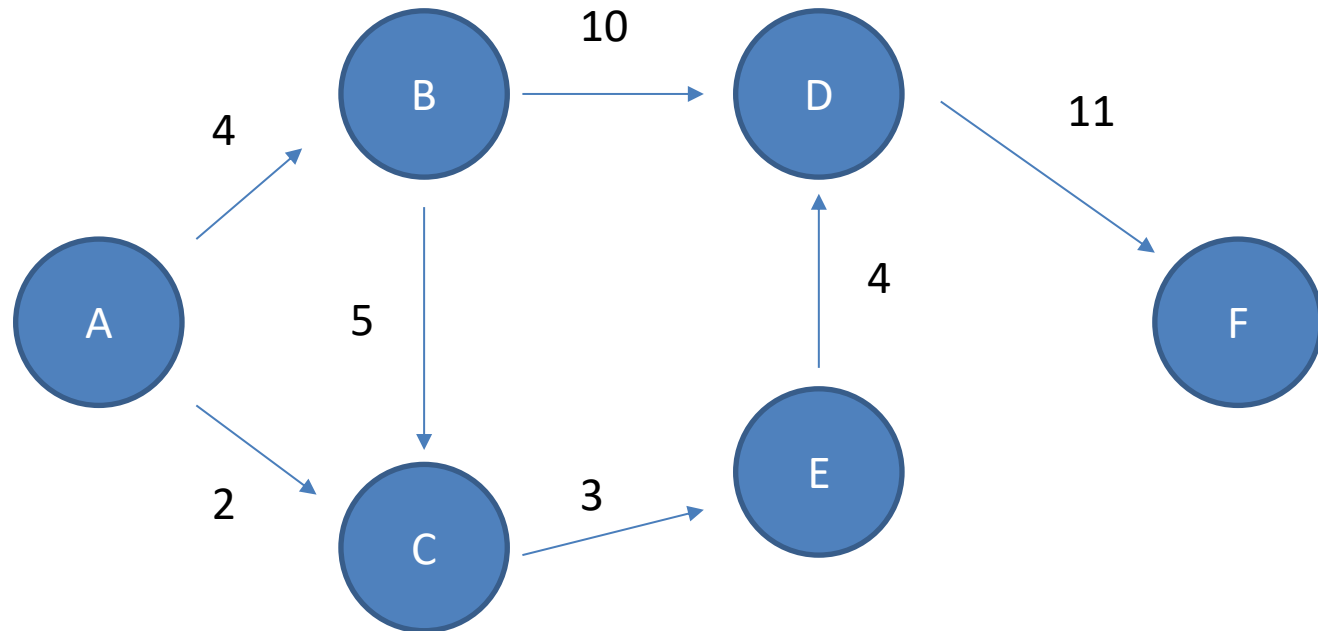
- ✦ Put elements of a list in an ascending order



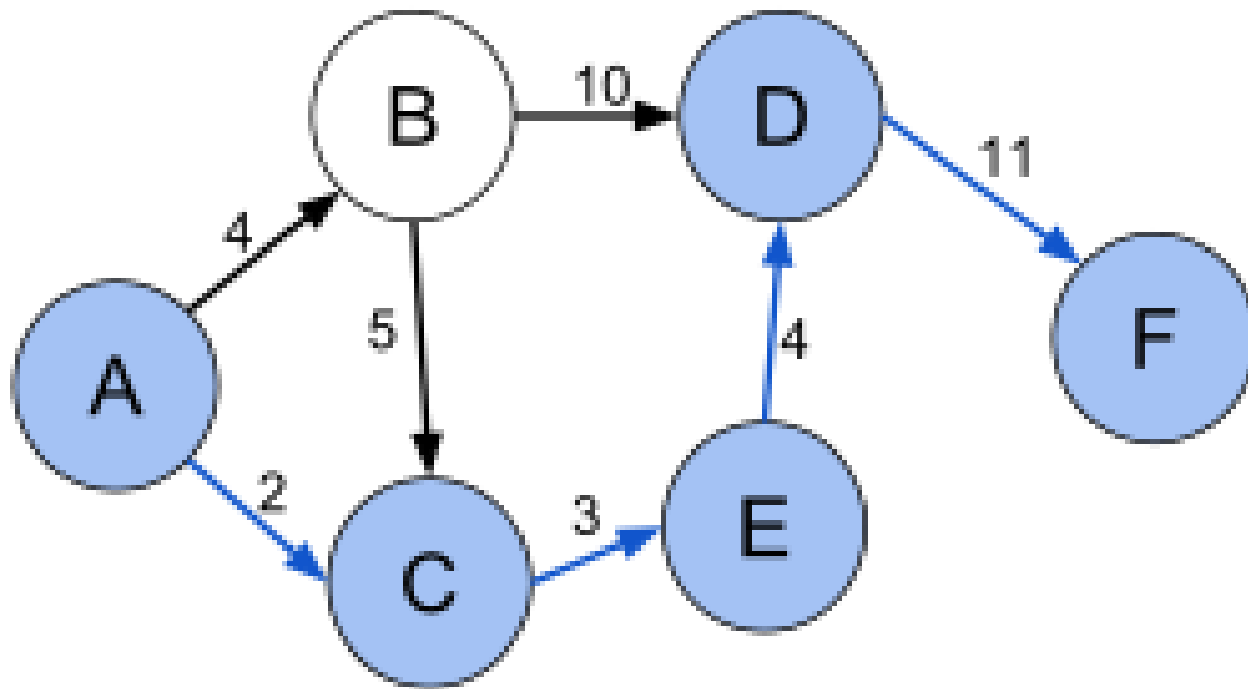
Some Basic Examples of C Programming

◆ Shortest path problem

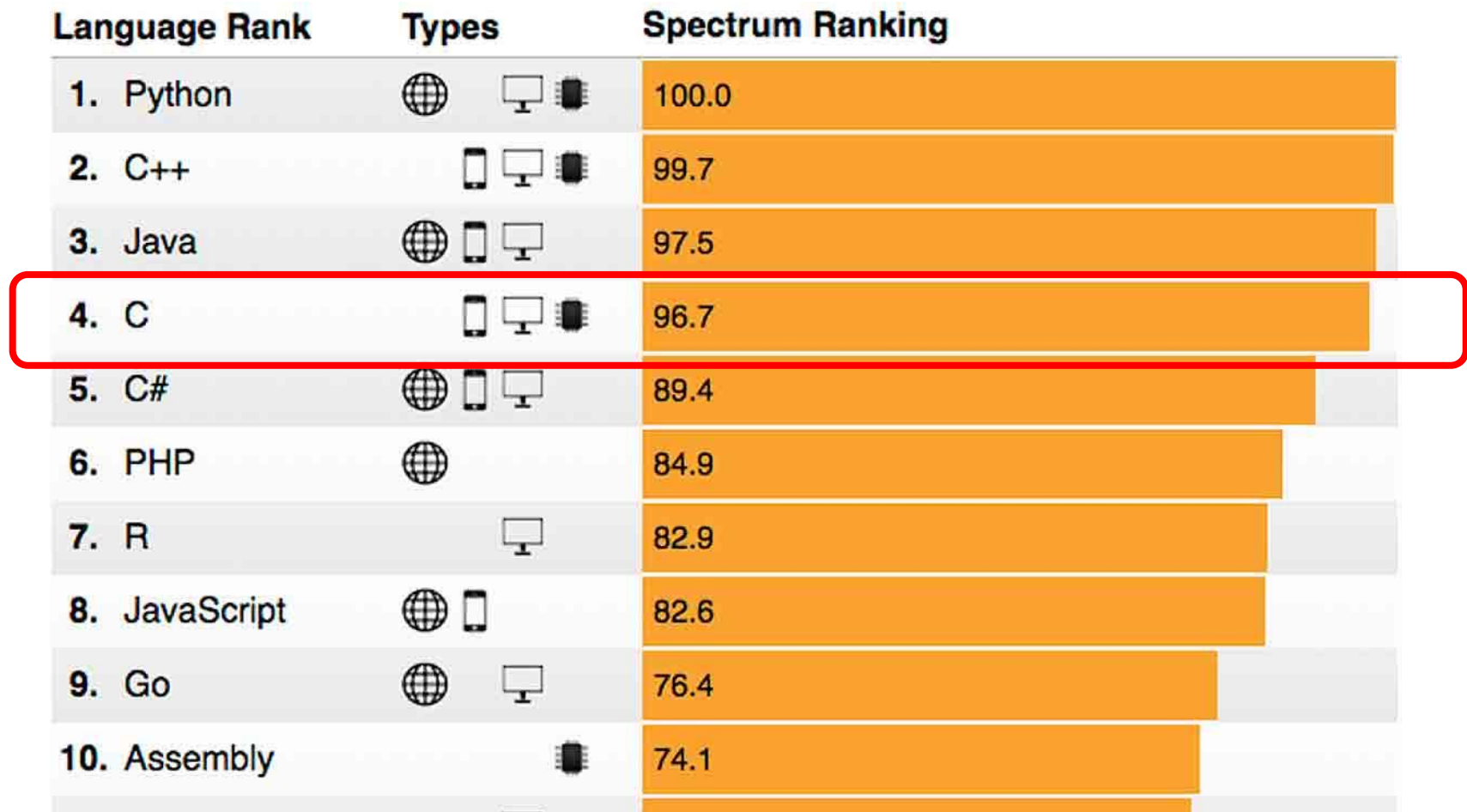
- ✦ Find a path between two nodes in a graph such that the sum of the weights of its constituent edges is minimized



Shortest path problem



Programming Languages



<https://spectrum.ieee.org/static/interactive-the-top-programming-languages-2018>

Programming Languages

◆ Python

- ❖ A scripting language that is often used by software developers to add programmability to their applications, such as engineering-analysis tools or animation software.

◆ C++

- ❖ Essentially a version of C with built-in support for "objects"--self-contained modules of code and data--C++ proved to be a natural fit for software driven by graphical user interfaces.

◆ Java

- ❖ Designed to allow the creation of programs that can run on different platforms with little or no modification, Java is a popular choice for Web applications.

◆ C

- ❖ C is used to write software where speed and flexibility is important, such as in embedded systems or high-performance computing.
- ❖ C was originally developed at Bell Labs by Dennis Ritchie, between 1972 and 1973.

The Basics of C Programming

◆ C

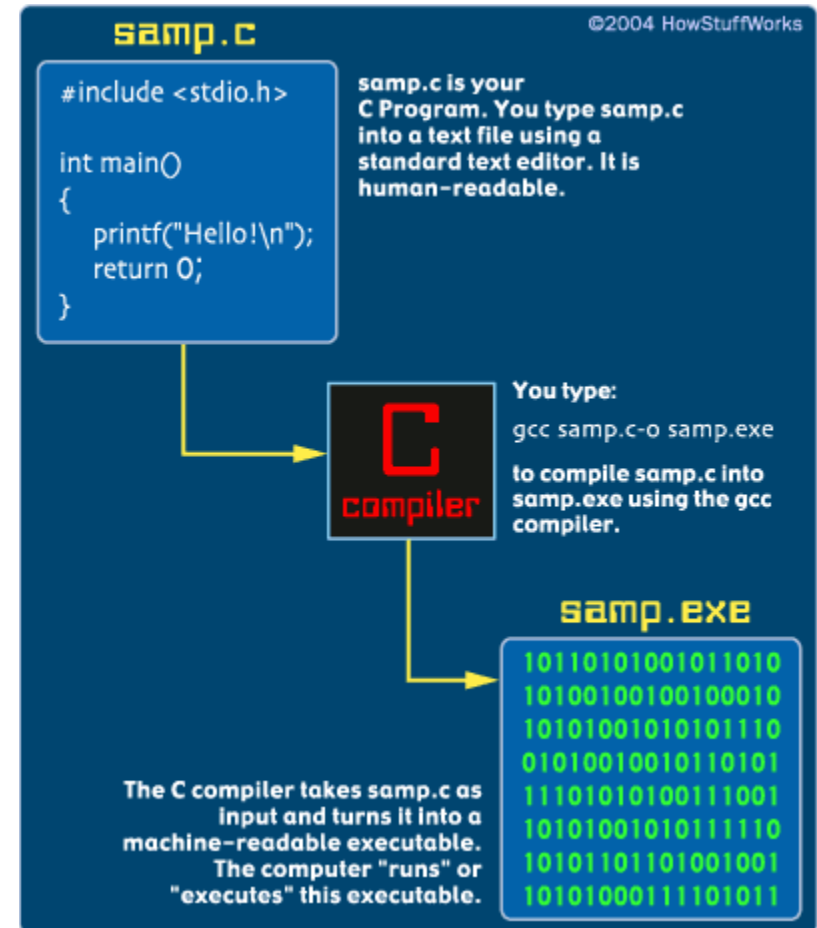
- ❖ Programming codes
- ❖ Human readable

◆ Compiler

- ❖ gcc for Linux
- ❖ Visual studio for Windows

◆ Executable file

- ❖ Human un-readable
- ❖ Machine readable



<https://computer.howstuffworks.com/c1.htm>

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Linux

- ◆ Free and open-source software operating systems based on the Linux kernel
 - ❖ Multi-user
 - ❖ Multi-task



Which is the symbol for linux?

Linux

◆ CentOS

❖ A linux distribution

◆ Log-in your linux

❖ ID: student id

❖ Password: course navi password



Keep the size of your files less than 1Gb

Open the Terminal



Linux Command

◆ pwd →

- ❖ Print name of current/working directory

◆ ls

- ❖ List directory contents

◆ mkdir →

- ❖ Make directories

◆ cd

- ❖ Change directory

◆ cp

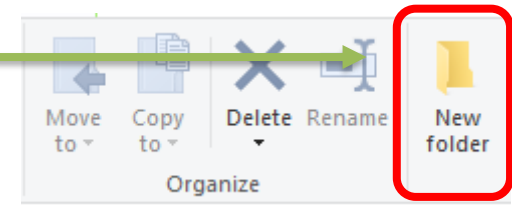
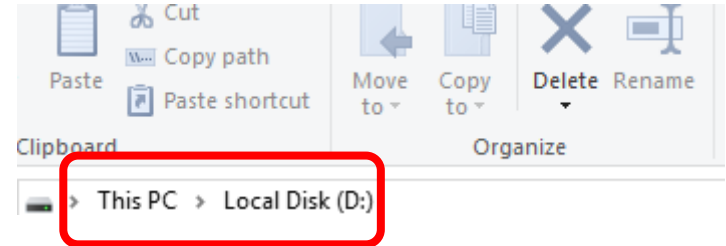
- ❖ Copy Files and directories

◆ mv

- ❖ Move (rename) files

◆ rm

- ❖ Remove files or directories



pwd: Print name of current/working directory

```
pwd
```

```
$ pwd ↵  
/var/autofs/homes/Linux/home01/1g04a****
```

⇒ Use this command to find out what directory you are working in.
Client Directory: the directory you are working in

✖ Home Directory: the directory you are in just after log-in

ls: List directory contents

ls [option] [name of directory]

\$ ls  (see what files and sub-directories you have)
Wnn7

\$ ls Wnn7  (see the directory that you specified)
address.h bunsetsu fisd.h muhenkan university.h

Option

\$ ls -F  (put / after the name of directories)
Wnn7/

\$ ls -a  (show all files)
.
..
.ICEauthority .adobe .gnome2
.fonts.cache-1 .gnome2_private
.gconf Wnn7

ls

Option

\$ ls -l 

(File Property)

合計 4

```
drwx-----  2 lg04a**** student      4096 2005-04-20 14:33 Wnn7
```

\$ ls -aF 

(Combination)

```
./          .adobe/      .gnome2/
../         .fonts.cache-1 .gnome2_private/
.ICEauthority .gconf/      Wnn7/
```

mkdir: Make directories

```
mkdir [name of directory]
```

```
$ mkdir work  (make the directory named by 「work」 )  
$ ls -F   
Wnn7/ work/
```

※ If there is no file in the directory, **rmdir** can be used to delete this directory

```
$ rmdir work 
```



How to delete non-empty folder?

cd: Change Directory

cd [name of directory]

\$ pwd  (confirm the directory before moving)
/var/autofs/homes/Linux/home01/1g09a****

\$ cd work  (move to work)

\$ pwd  (confirm the directory after moving)
/var/autofs/homes/Linux/home01/1g09a****/work

\$ cd ..  (move to the upper directory)

\$ pwd 
/var/autofs/homes/Linux/home01/1g09a****

✖ By only inputting

\$ cd 

you can move to the home directory.

Text editor

◆ An open source text editor

❖ Software to edit text and other applications

- gedit
- vim
- emacs

What is the meaning of “&”



```
$ gedit hello.c & ↵
```

❖ Create hello.c file and edit it by gedit

❖ You can also use vim or emacs according to your preference

gedit

```
#include <stdio.h>

int main(void)
{
    printf("hello, world¥n");
    return 0;
}
```

¥ could be \ (back slash)
according the computer
environment

⇒ Click "Save" icon to save the file

cp: Copy file

```
cp [File 1] [File 2]  
cp [File] [Directory]
```

```
$ ls
```

```
hello.c
```

```
$ cp hello.c goodbye.c
```

(Copy 「hello.c」 to
「goodbye.c」)

```
$ ls
```

```
goodbye.c hello.c
```

```
$ mkdir work2
```

```
$ cp hello.c work2
```

(Copy 「hello.c」 to
「directory work2」)

```
$ ls work2
```

```
hello.c
```

mv: Move file

```
mv [File 1] [File 2]  
mv [File ] [Directory]
```

```
$ ls -F
```

```
goodbye.c  hello.c  work2/
```

```
$ mv goodbye.c  byebye.c
```

(Move 「goodbye.c」 to
「byebye.c」)

```
$ ls -F
```

```
byebye.c  hello.c  work2/
```

```
$ mv byebye.c  work2
```

(Move 「byebye.c」 to
directory work2)

```
$ ls -F
```

```
hello.c  work2/
```

```
$ ls work2
```

```
byebye.c  hello.c
```

rm: Remove File

rm [Option] [File Name]

```
$ cd work2
```

```
$ ls
```

```
byebye.c  hello.c
```

```
$ rm -i byebye.c
```

```
rm: remove `byebye.c'? y
```

```
$ ls
```

```
hello.c
```

Option 「-i」 is to confirm the removing operation

Inputting y



will remove the file

Inputting n



will not remove the file

rm: Remove file

```
$ cd ..  
$ ls  
hello.c  work2  
$ rmdir work2 (rmdir can be only used when 「work2」 is empty)  
rmdir: `work2': this directory is not empty  
$ rm -r work2  
$ ls  
hello.c
```

rm with Option 「-r」 can remove the directory, which is not empty

Compile

Compile

GCC (GNU Compiler Collection)

GNOME :

```
gcc [Option] [File]
```

```
$ ls
```



```
hello.c
```

```
$ gcc hello.c
```



(Compile 「hello.c」)

```
$
```

⇒ If there is no error message, compiling is success
Otherwise, grammar mistakes exist

⇒ Revising the source file by Emacs
And compile it again

Execute

Execute

(a.out) will be created after the compiling
Run a.out at GNOME :

```
$ ls
```

```
a.out hello.c
```

```
$ ./a.out
```

```
hello, world
```

(Run 「a.out」)

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Exercise

- ◆ Print your student ID
 - ❖ “My student ID is YOURID”